Sikun Xu

CONTACT INFORMATION

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Olin Business School
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RESEARCH INTERESTS

Machine Learning and AI Systems; Causal Inference; Data-Driven Decision-Making; Recommendation Systems; LLM and Generative Models

EDUCATION

Washington University in St. Louis 2021–Present
- Ph.D. Candidate, Olin Business School
- Advisors: Raphael Thomadsen and Dennis Zhang

Columbia University in the City of New York 2019–2020
- M.S. in Operations Research

Shanghai Jiao Tong University 2015–2019
- B.S. in Industrial Engineering

JOB MARKET PAPERS

- The Winner's Curse in Data-Driven Decision Making: Evidence and Solutions. **Sikun Xu**, Raphael Thomadsen, and Dennis Zhang. *Submitted.*
- SOTA or Luck? The Winner's Curse in LLM Leaderboards. **Sikun Xu**, Raphael Thomadsen, Dennis Zhang, and Heng Zhang.

WORKING PAPERS

1. A Causal Approach to Representation Learning for Unstructured Data. **Sikun Xu**, Zhenling Jiang, Zhengling Qi, and Dennis Zhang. *Major revision at **Management Science**.*

CONFERENCE PROCEEDINGS

1. Optimal Real-Data Allocation for Synthetic-Data-Augmented Inference. **Sikun Xu**, Zineng Xu, and Mingduo Zhao. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. Forthcoming.
2. Verifying Global Optimality of Candidate Solutions to Polynomial Optimization Problems Using a Determinant Relaxation Hierarchy. **Sikun Xu**, Ruoyi Ma, Daniel K. Molzahn, Hassan Hijazi, and Cédric Josz. *Proceedings of the 60th IEEE Conference on Decision and Control (CDC)*, pp. 3143–3148, 2021.

INDUSTRY EXPERIENCE

- **TikTok Inc.**, San Jose, Machine Learning Scientist Intern 2026
- **Sauma Capital LLC**, New York, Quantitative Researcher 2021
- **Capstone Investment Advisors**, New York, Investment Intern 2020
- **Cardinal Operations**, Shanghai, Algorithm Engineer Intern 2018

INVITED SEMINARS

- Washington University in St. Louis, Department of Statistics and Data Science 2026
- The Chinese University of Hong Kong, Business School 2025

CONFERENCE PRESENTATIONS

SOTA or Luck? The Winner's Curse in LLM Leaderboards

- Conference on Information Systems and Technology, CA (Scheduled) 2026

The Winner's Curse in Data-Driven Decision Making: Evidence and Solutions

- INFORMS Annual Meeting, San Francisco, CA (Scheduled) 2026
- INFORMS Annual Meeting, Atlanta, GA 2025
- ISMS Marketing Science Conference, Washington, D.C. 2025
- Conference on Artificial Intelligence, Machine Learning, and Business Analytics, New Haven, CT 2024

A Causal Approach to Representation Learning for Unstructured Data

- ISMS Marketing Science Conference, Washington, D.C. 2025
- INFORMS Annual Meeting, Seattle, WA 2024
- POMS Annual Conference, Minneapolis, MN 2024
- INFORMS Annual Meeting, Phoenix, AZ 2023

Data-Driven Security Selection for Wealth Management

- POMS Annual Conference, Orlando, FL 2023
- INFORMS Annual Meeting, Indianapolis, IN 2022

Verifying Global Optimality of Candidate Solutions to Polynomial Optimization Problems Using a Determinant Relaxation Hierarchy

- INFORMS Annual Meeting, Anaheim, CA 2021

TEACHING EXPERIENCE

Guest Lecturer

- MGT 680E AI & Machine Learning Business Applications, Washington University in St. Louis Fall 2024
- IEOR 4721 AI Application in Finance, Columbia University Spring 2022
- IEOR 4721 AI Application in Finance, Columbia University Summer 2021

Teaching Assistant

- Olin Business School, Washington University in St. Louis
  - MGT 680E AI & Machine Learning Business Applications
  - SCOT 519E Revenue Management
  - SCOT 500M Supply Chain Analytics: Stochastic Models
  - SCOT 558 Advanced Operations Strategy
  - SCOT 5704 Operations Management
  - SCOT 356 Operations and Manufacturing Management
  - SCOT 400D Supply Chain Management
  - SCOT 500D Project Management
- Columbia University in the City of New York
  - IEOR 4742 Deep Learning
  - IEOR 4525 Machine Learning

ACADEMIC SERVICE

- Reviewer: *Journal of Investment Strategies*, *Conference on Information Systems and Technology (CIST)*
- Session Chair: INFORMS Annual Meeting, CIST

REFERENCES

Professor Raphael Thomadsen
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The Wharton School
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